

# Which stocks could we really have studied?

Daily learning note | 25 September 2026 | Q004

Sidiq review status: Not yet reviewed.

## Research question and hypothesis

Can a data source support historical stock selection, return accounting and information timing under our actual access permissions? The hypothesis is that advertised inactive-stock coverage is insufficient: unknown access, missing terminal-event evidence or unavailable historical vintages must block admission.

Q003 identified the bytes behind a result. Q004 asks whether those bytes would describe the right population and whether we can lawfully obtain and use them. Today completes a universe specification and source comparison, with a small evidence checker and synthetic diagnostics. It acquires no real market dataset.

## Beginner explanation and terms

Imagine studying a school using only the pupils still enrolled at graduation. Those who left are missing from the earlier classes too. A current stock list can make the same mistake: companies that disappeared are excluded from the history even though they were available to buy at the time.

**Universe:** the securities eligible for consideration at a decision time. **Survivorship bias:** selection based on later survival. **Delisting:** removal from a trading venue, not necessarily economic worthlessness. **Permanent identifier:** a security identity that survives ticker changes. **Vintage:** the version of data available at a particular time. **Entitlement:** the actual permission to access a product for a stated use.

Historical coverage and historical availability are separate. A corrected 2012 price retrieved today is an observation about 2012, but that corrected value may not have been available during 2012. An accurate checksum of today's file cannot prove an old trader knew its contents.

## Data, metrics and protocol declared before implementation

Source evidence is public documentation consulted today, recorded separately from invented test inputs. The proposed equity study window is 2010-2025, with earlier warm-up sessions; no market observations from that window are downloaded or tested. The return example covers one invented holding period with three holdings and no historical dates. There is no fitted model, tuning procedure or observed alpha.

Acceptance metrics: zero admissions for the two unverified candidates; all eight omitted gates invalid; 24 blocked single-gate downgrades; exact repeatability of the result JSON; and hand-checked sample returns. An all-verified synthetic record is the positive control, not an assertion that we possess licensed data.

## Provisional universe and mathematics

The first price-based study targets USD-quoted common shares of operating/holding companies on NYSE, Nasdaq and NYSE American, including REIT common shares. Qualifying share classes retain separate permanent security IDs. ADRs, funds, preferreds, units, partnerships, warrants, blank-check companies and OTC-only listings are excluded. Historical classification is required; today's classification cannot be projected back.

Decision time is provisionally 09:00 America/New\_York on a trading session. Use only available prior-session inputs. The prior raw close must be at least USD 5. Mean close times share volume over the 20 prior sessions must be at least USD 1 million per session. Missing inputs exclude a new entry with a recorded reason. These are untuned teaching defaults, not a claim about executable liquidity or strategy profitability.

Let  $H_i(t)$  describe eligible historical listing/type and  $A_i(t)$  require information available by  $t$ . Let  $L_i(t) = \text{sum}(P_i(s) * V_i(s) \text{ over } 20 \text{ prior sessions } s) / 20$ . Then eligibility is  $E_i(t) = H_i(t) \text{ AND } A_i(t) \text{ AND } [P_i(d-1) \geq 5] \text{ AND } [L_i(t) \geq 1000000]$ .  $P$  is unadjusted USD close and  $V$  is consolidated shares traded; their product is a turnover proxy. All inputs must be known. Eligibility loss does not erase an existing holding or establish its exit price.

The complete rule is in data/universe-and-access.md. Q006 will implement the calendar and decision-time contract; Q009 will implement membership. These components and executable portfolio rules are not built today.

## Worked example: three initial holdings

Each synthetic holding starts with 100 units. There are no intervening cash flows or costs. C's zero is an explicitly confirmed zero recovery in the fixture, never a default for a missing real price.

| Holding | Initial value | Final value | Later status                    |
|---------|---------------|-------------|---------------------------------|
| A       | 100           | 120         | Survives                        |
| B       | 100           | 90          | Survives                        |
| C       | 100           | 0           | Disappears after confirmed loss |

The full population return is  $(120 + 90 + 0) / 300 - 1 = -0.30$ , or -30%. A survivors-only history reports  $(120 + 90) / 200 - 1 = 0.05$ , or +5%. The selection difference is  $0.05 - (-0.30) = 0.35$ , or 35 percentage points.

This comparison changes the population and initial capital consistently. It demonstrates selection error, not its empirical size. An acquired stock can leave with a gain, so the sign need not always be positive.

## Sources, access and revisions

CRSP's official product page documents active/inactive coverage, permanent identifiers and corporate actions. The CRSPAccess field guide distinguishes terminal proceeds and missing delisting returns. The newer delivered format must be checked separately; never assume a legacy field convention applies unchanged. <https://indexes.morningstar.com/research-data-products/crsp-us-stock-databases>  
[https://www.crsp.org/crsp\\_pdf/crsp-us-stock-indexes-databases-data-descriptions-guide-crspaccess/](https://www.crsp.org/crsp_pdf/crsp-us-stock-indexes-databases-data-descriptions-guide-crspaccess/)

Norgate's subscription/content pages identify Platinum inactive-stock coverage and distinguish OTC migration from no-longer-tradeable securities. Its access overview requires Windows or a Windows VM. The data FAQ says historical corrections are applied without versioning, so current histories alone cannot satisfy our strict vintage requirement. Both entitlement and usable local access remain unestablished for this project. <https://norgatedata.com/prices.php>  
<https://norgatedata.com/data-content-tables.php> <https://norgatedata.com/>  
<https://norgatedata.com/data-package-faq.php>

Norgate's EULA sections 8 and 21 restrict redistribution and require content deletion on subscription expiry, with retention allowed for defined derived data. Our decision is to keep vendor content out of Git and establish the actual storage/output permissions before access. Public documentation grants no entitlement. <https://norgatedata.com/subscribe/eula.php>

These primary sources were consulted on 25 September 2026. Candidate suitability is our inference from public documentation; no licensed sample has been audited. No purchase, trial or download was initiated.

## Implementation and checks

The reusable `src/quant_research/data/access.py` checker requires eight gates: licence entitlement, local access, historical identity, inactive coverage, corporate actions, terminal events, availability vintages and permitted outputs. Each has a status and a nonblank evidence reference. Unknown fields, omissions, malformed records and invalid statuses fail validation. Blocker order is deterministic.

Only verified passes. Documented is a vendor claim; unknown is unresolved; failed records contradictory evidence for this contract. These three statuses all block. The two dated real-candidate records contain no verified gates. The checklist checks declarations; it does not authenticate references or permissions, fetch data, enforce ingestion or validate security membership. A dishonest declaration can pass it.

Run `make access-experiment` to reproduce the dated candidate assessment, positive control and all negative controls. Run `make check` for the full suite. Run `make note NOTE_DATE=2026-09-25` to rebuild this PDF. Tests compare the arithmetic to literal expected values and compare the result file byte-for-byte.

## Observed results and validation

| Check                          | Observed result                                                    |
|--------------------------------|--------------------------------------------------------------------|
| Real-source admission          | 0 of 2 candidates ready; 8 unverified gates each                   |
| Synthetic complete declaration | Accepted as a checklist positive control                           |
| Single-gate downgrades         | All 24 blocked; no vendor statement treated as verification        |
| Synthetic return example       | Full population -30%; survivors +5%; difference 35 points          |
| Python tests                   | 144 passed, including 50 new access cases                          |
| Code quality                   | Ruff passed; 25 files formatted; mypy passed for 15 source files   |
| C++ and dependencies           | Warning-clean build, expected output and pip consistency passed    |
| Repeatability                  | Two experiment runs matched each other and the stored JSON exactly |

The first full check found one overlong string in the experiment; splitting the literal corrected lint without changing the output. The initial restricted fetch failed DNS; the approved fetch succeeded. The repository began clean and synchronized, and the authenticated issue read returned no open issues. No coverage percentage or remote CI outcome is inferred from local checks.

## Limitations and interview explanation

No source is empirically admitted, no market coverage is measured, and no live investment recommendation is made. A hash proves neither licensing nor historical completeness. Today's checker reports human evidence assertions; it is not a permission service or an ingestion barrier. Future code must enforce its own contracts. Strict historical receipt/vintage requirements may require a different source or a documented narrower claim.

In an interview, explain the changing population before describing returns. Show the three-holding arithmetic, then distinguish a missing price from a confirmed loss. Explain why inactive coverage, historical membership, revision timing and permission are separate requirements. Finish with what the implementation actually checks.

## Next work and required input

Q005a is next: fix named-zone fold comparisons before general ingestion. Q004a remains open for lawful sample acquisition and audit. Before that step, Sidiq should identify existing licensed access or decide a source/budget; credentials should not be sent. Synthetic engineering continues without that answer. C002 remains the next Wednesday step, 30 September. Personal review remains Not yet reviewed.

## Check your understanding

1. Why can a current stock list distort an older backtest?
2. Does a delisting mean a -100% return?
3. Why does documented source coverage fail the admission checklist?
4. Can a one-day lag repair a historical revision whose original version is unavailable?

## Suggested answers

1. It conditions the past sample on later survival, excluding securities that were then eligible. Our toy sample changes from -30% to +5%, but real selection effects need not have that magnitude or sign.
2. No. A company may migrate venue or be acquired. A terminal return requires proceeds and payment evidence; missing proceeds are unresolved, whereas the fixture explicitly specifies zero recovery.
3. A vendor statement establishes neither actual permission nor successful sample verification in this project. Every requirement needs project evidence; a nonblank reference alone does not establish truth.
4. No. Delaying a revised value does not recover the missing original vintage. The claim must remain blocked or be explicitly narrowed before research using the revised history.